

BRYCE ROBERTS

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bryceroberts13.github.io

Education

B.S. in Artificial Intelligence & Decision Making, Mathematics (Double Major) Aug. 2023 – May 2027
Massachusetts Institute of Technology **GPA: 4.6/5.0**

Relevant Coursework: Design and Analysis of Algorithms, Machine Learning (Graduate), Non-Linear Optimization (Graduate), Statistics, Discrete Probability and Stochastic Processes (Graduate), Theory of Computation, Distributed Algorithms, Machine Learning, Introduction to Programming, Linear Algebra, Representation, Inference, and Reasoning in AI, Strategy and Information, Mathematical Economic Modeling, Algebraic Combinatorics

Experience

General Intelligence Company

Fall 2025

Research Intern

New York, NY (Remote)

- Developing novel benchmarks to evaluate long-context memory capabilities in large language models (LLMs).
- Designing and implementing experiments to gauge and improve LLM performance on extended memory tasks.

Machine Learning Research — Spoof Detection

Spring 2025

Researcher / Collaborator

Massachusetts Institute of Technology

- Researching spoof detection using CLIP embeddings with contrastive/triplet learning, optimizing similarity metrics and evaluations, while assisting in writing and preparing a research paper for publication.

Contraca

January 2025 – August 2025

Software Engineer

Boston, Massachusetts

- Designed and implemented backend architecture and integrated APIs to enable scalable, dynamic content generation and distribution for the core product, as well as integration with API to deliver front end interactivity.
- Developed web scrapers to collect detailed external data, informing automated content creation pipelines.
- Engaged directly with clients to translate feedback into product improvements and represented the company at industry conventions to gather market insights and track emerging technologies.

Tier1Alpha

July 2023 – August 2023

Research Intern

San Francisco, California

- Researched and wrote financial blogs explaining key concepts and options trading strategies for clients.
- Automated parts of the blog creation process, improving efficiency and tailoring content delivery to client needs.

Leadership / Extracurricular

Varsity Swim

September 2023 – Present

Member

Massachusetts Institute of Technology

- Won the 100m Butterfly at the **NEWMAC Championship**, setting a conference record.
- Awarded **Rookie Athlete of the Year** for exceptional performance in debut season.
- Competed in four relay events and two individual events at **NCAAs**, contributing to the team's performance.

Ring Committee

Spring 2024 – Spring 2025

Artist and Study Break Chair

Massachusetts Institute of Technology

- Worked with a **12-member committee** to design **MIT's class ring** to reflect the class's values and experiences.
- Planned and lead the **Study Break event** to gather class input and shape the final ring design.
- Coordinated communications with vendors and committees to successfully execute social events for the ring design presentation and ring delivery.

Technical Skills

Languages: Python, HTML/CSS, TypeScript, Bash Scripting, C, AppleScript

Developer Tools: Terminal, VS Code, Jupyter Notebooks, Git, Github, UV

Libraries/Frameworks: NumPy, Pandas, Matplotlib, Scipy, PyTorch, Selenium with Python, BeautifulSoup

Projects

Targeted Lobbying on Council Networks | *MIT 14.18 Mathematical Economic Modeling* Spring 2026

- Coauthored final paper modeling a budget-constrained lobbyist on a legislature with network strategic complementarities; passage institution maps members' actions into adoption probability.
- Under linear aggregation of support, cost minimization spikes spending on highest Katz–Bonacich central member; under smooth strictly increasing passage, small-budget optimum targets slope-weighted Bonacich index (spillovers times marginal vote responsiveness at no-lobbying equilibrium).
- Embedded linear–quadratic network game (Galeotti–Golub–Goyal style), worked closed-form small-network examples, and extensions (bounded actions with nonnegative spillovers; probabilistic-vote companion).

Custom Keyboard Optimization | *QMK, Karabiner, Bash, AppleScript* Summer 2024

- Implemented a leader key with programmable scripts and custom key remappings to optimize workflow efficiency.
- Researched and applied QMK firmware, Karabiner-Elements, Bash, and AppleScript for deep system customization.

Personal Portfolio Website | *React.js, CSS* | bryceroberts13.github.io June 2025

- Developed a responsive React.js portfolio site featuring a modern UI to showcase projects and professional experience.
- Designed and implemented interactive components to effectively present technical work and background.

Fractional Coloring Presentation | [View Presentation](#) February 2023

- Conducted research on fractional coloring, proving multiple theorems about the fractional chromatic number of a special class of graphs, and presented findings in a clear and organized presentation.

Sepsis Early Detection with Machine Learning | *MIT 6.790 Graduate Machine Learning* December 2024

- Developed predictive models on ICU patient time-series data to detect sepsis up to 12 hours before clinical diagnosis.
- Implemented missing data strategies (median and regression imputation), class imbalance handling (oversampling), and utility-aware loss functions to optimize clinically relevant predictions.
- Compared logistic regression and recurrent neural networks (RNNs); achieved **82% accuracy** with an RNN using regression imputation, oversampling, and weighted loss, outperforming baseline methods.

Instagram Web Scraper June 2023

- Developed a web scraper to automate the retrieval and storage of friends' Instagram posts.
- Implemented scraping techniques to bypass dynamic content loading, ensuring efficient data collection.
- Utilized Python libraries such as Selenium and BeautifulSoup to navigate and extract structured data.